

Solid SD Preparation

Solid medium allows for the selection of individual colonies. This protocol makes about 20 plates from 500mL of liquid synthetic defined / synthetic dextrose (SD) media deficient in uracil.

Reagents

✓	Name	Quantity	Location/Notes
☐	Yeast Synthetic Drop-out Medium Supplement	0.46g	LACKS URACIL or LEUCINE
☐	Yeast Nitrogen Base without amino acids	3.35g	
☐	ddH ₂ O	500mL	
☐	D-glucose / dextrose	10g	
☐	Agar powder	10g	
☐	Sterile water	As needed	

Equipment

✓	Name	Quantity	Location/Notes
☐	Analytical scale		
☐	Autoclave		Training required

☐	500mL autoclaveable bottle		
☐	1L autoclavable bottle		
☐	Sterile Agar plates	20	
☐	Parafilm		
☐	Sterile bags		
☐	Stir bar		
☐	BSC		Training required

Instructions

1. Prepare the dextrose solution by adding the following to the **500mL bottle**, use a stir bar to stir:

Reagent Tracker	Quantity
☐ D-glucose / dextrose	10g
☐ ddH ₂ O	to 100mL

2. Prepare the rest of the solution by adding the following to the **1L bottle**, use a stir bar to stir:

Reagent Tracker	Quantity
☐ Yeast Synthetic Drop-out Medium Supplement	0.46g

☐ Yeast Nitrogen Base without amino acids	3.35g
☐ Agar powder	10g
☐ ddH ₂ O	to 400 µl

3. Autoclave both liquids at 121°C for 15min and allow them to cool to about 50°C
4. In a BSC, unstack sterile plates
5. Add the dextrose solution to the 1L bottle, add sterile water until the solution is 1L
6. Pour the media into the plates and allow to cool until solid (30min - 1h)
7. Seal plates with parafilm, label, and place in sterile bags to store at 4°C

Notes here

1. Store upside-down to reduce condensation
2. DO NOT RE-AUTOClave, can try to use a water bath to re-melt solidified agar (see below)
3. Solutions must be autoclaved separately and later combined to avoid triggering the maillard reaction in the autoclave , which could make toxic by-products and lower glucose concentration [4]

References

[1]

www.sigmaaldrich.com/CA/en/technical-documents/protocol/microbiological-testing/microbial-culture-media-preparation/yeast-drop-out#products

[2]

<https://www.protocols.io/view/yeast-peptone-dextrose-ypd-medium-x54v9d7xlq3e/v1>

[3]

https://bio.libretexts.org/Bookshelves/Biotechnology/Lab_Manual%3A_Introduction_to_Biotechnology/01%3A_Techniques/1.19%3A_Pouring_Agar_Plates

[4]

https://www.researchgate.net/post/Autoclaving_Glucose_in_yeast_peptone_Dextrose_plate_with_Agar

[5] <https://bitesizebio.com/32884/pour-agar-plates-pinch/>